

Aft-Fan Inflow Induction and Wing Separation During Tail-Sitter Transition

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A tail-sitter with an aft-mounted ducted fan spends its transition at incidences where the wing is separated, and the fan sits close enough behind the wing to change that. The mechanism is not the one a blown-wing study would model: the duct leading edge lies a fifth of a chord *aft* of the wing trailing edge, so its slipstream never reaches the wing. What reaches the wing is the fan’s *upstream induction* — the accelerating field ahead of a loaded disk — which is a favourable streamwise pressure gradient and therefore delays separation. This paper models that induction as the semi-infinite cylindrical vortex sheet equivalent to a uniformly loaded actuator disk, verified against the closed-form axial solution and against momentum theory, and applies it as an external field to a coupled wing–body panel solve whose separation behaviour is established independently. On the demonstrator airframe the fan is found to buy $\Delta\alpha \approx 0.8^\circ$ of incidence at the transition point, measured as the shift of the separation-location curve rather than as the crossing of a single threshold — a distinction that matters, since a threshold evaluated on a two-degree grid reports no effect at

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all from data containing this one. The benefit is concentrated inboard, over precisely the span station that separates first, and reverses sign outboard where the wing leaves the fan’s streamtube and meets its return flow.

Nomenclature

A	disk area, m ²
b	wing span, m
c_n	chord in the leading-edge-normal plane, m
C_L	lift coefficient
R	disk radius, m
s	axial distance from the disk plane, m
T	fan thrust, N
u_x	axial induced velocity, m/s
v_h	hover induced velocity, $\sqrt{T/(2\rho A)}$, m/s
v_i	induced velocity at the disk plane, m/s
V_∞	freestream speed, m/s
x_{sep}	chordwise separation station, fraction of c_n
α	angle of attack, deg
γ_t	cylindrical vortex sheet strength, m/s
μ	velocity ratio, V_∞/v_h
ρ	density, kg/m ³

I. Introduction

Transition is the part of a tail-sitter’s envelope that sizes its wing and its control authority, and it is the part a potential-flow method is least able to speak to, because the wing is separated over much of it. The companion study of this airframe establishes where separation begins and how far forward it moves with incidence. This paper asks what the propulsion system does about it.

The question is usually posed for a *tractor* installation, where the propeller slipstream washes the wing and raises its dynamic pressure — a blown wing, extensively studied. The airframe here is the other arrangement, and the difference is not a detail. Its ducted fan sits at the tail: the duct leading edge is 0.21 chord *behind* the wing trailing edge, so the slipstream leaves downstream and never touches the wing. A model that represents only the wake therefore reports exactly zero interaction, which is a statement about the model rather

than about the aircraft.

What does reach the wing is the induction *ahead* of the disk. A loaded actuator disk accelerates the flow approaching it, from zero far upstream to v_i at the disk plane; a wing sitting in that field experiences a favourable streamwise pressure gradient, and a favourable gradient delays separation. The effect is the same in sign as a blown wing’s and entirely different in mechanism, and it is much less studied.

Two features of this configuration make the question sharp rather than academic. First, the induced velocity is not a small correction: with the annular disk area of the demonstrator, v_h runs 14–24 m/s against a 25 m/s cruise, so the induction is comparable to the flight speed throughout transition and largest exactly where the airspeed is lowest. Second, the geometry aligns: the duct outer radius is 0.21 semi-span, and the companion study places the wing’s most advanced separation at $|2y/b| \approx 0.15$ –0.23. The part of the wing that sheds first is the part the fan protects.

II. The induction model

A. Why the wake model cannot serve

The solver’s existing slipstream model reconstructs a rotor’s wake from annular momentum theory and returns zero induced velocity at any point ahead of the disk plane. That is a correct economy for a surface lying in the wake, and it is the wrong tool here by construction. Applied to a wing upstream of the disk it returns zero, and the zero is the model’s rather than the flow’s.

B. The disk as a vortex cylinder

A uniformly loaded actuator disk is exactly equivalent to a semi-infinite cylindrical vortex sheet: azimuthal vorticity of constant strength $\gamma_t = 2v_i$ per unit length, shed from the disk edge and trailing downstream. The equivalence is what makes the upstream field tractable, because it turns “what does a disk do ahead of itself” into a Biot–Savart integral over a known geometry with a closed-form answer on the axis:

$$u_x(s) = \frac{\gamma_t}{2} \left[1 + \frac{s}{\sqrt{s^2 + R^2}} \right], \quad (1)$$

with s measured downstream from the disk plane. Equation (1) returns v_i at the disk, $2v_i$ far downstream and zero far upstream — momentum theory’s three values, recovered from the vortex system rather than assumed — and $v_i(1 - 1/\sqrt{2}) = 0.293 v_i$ one radius upstream, which is the number the whole interaction argument turns on.

Off the axis the sheet is discretized into vortex rings, each polygonized into straight filaments and evaluated with the same Biot–Savart kernel the lattice uses for its own wake. Ring spacing grows geometrically downstream, since the first radius or two sets essentially the entire upstream answer while the far tail varies slowly, and that spacing keys to each cylinder’s own radius rather than the disk’s outer one.

A ducted fan around a tail boom is an *annulus*, not a disk. A uniformly loaded annulus carries no trailing vorticity between its radii, so it sheds only at the two edges — the outer at $+\gamma_t$, the inner at $-\gamma_t$ — and is therefore two superposed cylinders. On the axis the two cancel at the disk plane and asymptotically but not in between: downstream inside the bore the inner sheet dominates and the axial induction reverses, which is the centerbody wake and a real feature of annular jets.

Swirl is deliberately absent. The root vortex and the wake’s axial vorticity lie downstream of the disk and contribute nothing ahead of it, which is the region this model exists to serve.

C. Verification

The discretized sheet is checked against Eq. (1) at stations from four radii upstream to ten downstream, holding to 2% of v_i , with the transverse components vanishing on the axis by symmetry and the error falling under refinement. The annulus is checked against its own closed form, which pins the inner sheet’s sign and strength, including the reversed induction inside the bore. The momentum inversion $T = 2\rho A v_i (V_\infty + v_i)$ round-trips, and forward speed reduces v_i at fixed thrust.

One further check is the one that magnitude comparisons cannot make: the induction must point *downstream* for a thrusting disk, both in the jet and ahead of it, and must reverse when the disk axis reverses. Inverting the ring circulation sense would turn the upstream acceleration this paper rests on into a deceleration while leaving every magnitude correct.

III. Application to the demonstrator

The induction is applied as an external perturbation field to the coupled wing–body–duct solve of the companion study, and the separation march is run with the fan on and off at otherwise identical conditions. The coupling is presently *one-way*: the airframe sees the fan, the fan does not see the airframe, so every result below is a lower bound on the interaction.

A. Magnitude and distribution

The quantity that matters is not the induction at a point but its variation *along the chord*, because that variation is the pressure gradient. The wing trailing edge sits roughly three

times closer to the disk than its leading edge, so the induction rises substantially across the chord: at the transition point $V_\infty = 12$ m/s, $T = 25$ N, the rise is 23% of freestream at $2y/b = 0.10$, 14.5% at 0.20 and 5% at 0.30. It is a favourable gradient, and it is strongest at the trailing edge, which is where turbulent separation begins and from which it marches forward.

Two features of the spanwise distribution deserve emphasis. It falls by a factor of four between $2y/b = 0.10$ and 0.30, concentrating the benefit inboard — over the stations that separate first. And beyond $2y/b \approx 0.4$ it *reverses*: outside the vortex cylinder the induced axial velocity is a return flow, so the fan marginally penalizes the outer wing. The penalty is under 2% and the sign change is real, not numerical, and it exists in the answer only because the upstream field is modelled at all.

B. How much incidence the fan buys

Measuring the benefit requires care, and the obvious measurement fails. Defining separation onset as a threshold crossing — the first α at which any station separates forward of a fixed chordwise station — and evaluating it on the two-degree grid used for the coefficient tables returns *no change at all* between fan on and fan off. That result is an artifact: a shift smaller than the grid spacing moves the crossing within an interval without moving it across one, so the reported onset is quantized to the grid and any sub-degree shift is invisible by construction.

The appropriate measurement is the horizontal shift of the separation curve $x_{\text{sep}}(\alpha)$, evaluated at several criterion levels. Measured that way on a 0.5° grid, the fan buys $\Delta\alpha = +0.81^\circ$ averaged over criteria from $x_{\text{sep}} = 0.85$ to 0.55, with individual values between $+0.73^\circ$ and $+0.86^\circ$. The consistency across criterion levels is what makes this a property of the flow rather than of the definition; the shift is smaller near onset, $+0.52^\circ$, where the separation point lies at the trailing edge and the layer is least sensitive to the gradient.

The benefit scales as expected with the velocity ratio: at cruise speed the same comparison returns roughly a third of the transition value, consistent with the induction scaling as v_i/V_∞ .

IV. Limitations

The coupling is one-way, so the reported shift is a lower bound. The induction model assumes uniform disk loading; real loading tapers at both edges, softening the sheet into a band that superposed cylinders at intermediate radii would represent. The fuselage boundary layer that an aft fan genuinely ingests is not modelled, and it acts to reduce the benefit. The analysis is quasi-steady, while transition is a manoeuvre rather than a sequence of trim

points. And the separation criterion inherits the companion study’s caveat: bubble bursting is not modelled, so the separation boundary is an upper bound on usable incidence rather than a predicted stall.

V. Conclusions

[Draft: conclusions pending the two-way coupling and the transition operating line. See the repository README.]

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